

Neuroinformatics Project

Classifying Neural Activity

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Course: Neuroinformatics

Date: 01.05.2025

Visualizing Firing Rate over Time

Mean Firing Rate over Time:

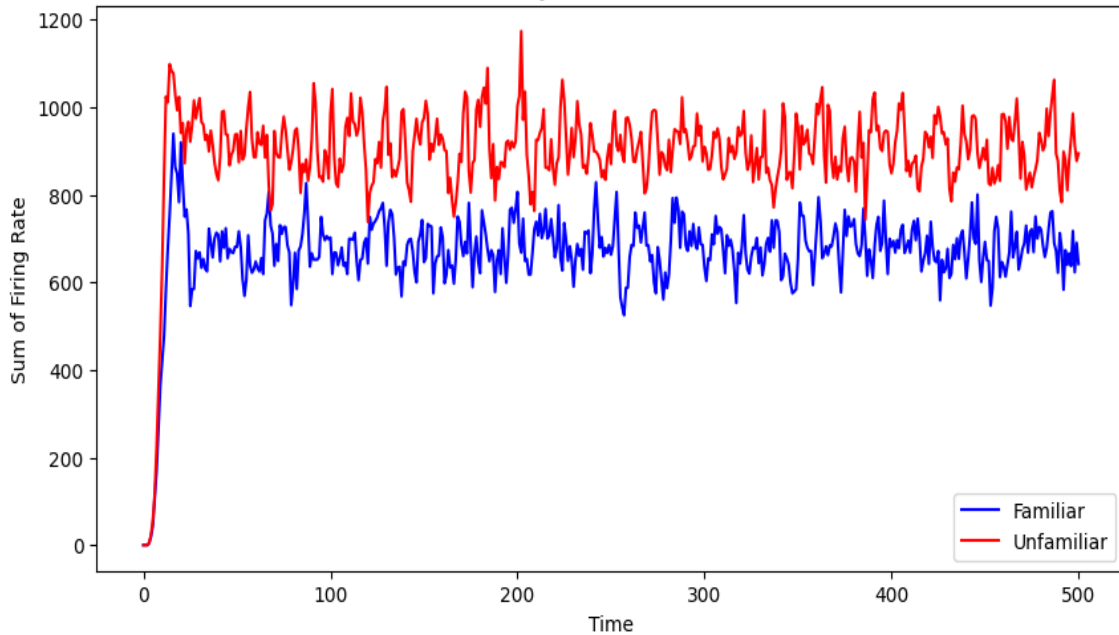
- Captures population response dynamics to stimuli.
- Helps distinguish familiar vs. unfamiliar stimuli.
- Provides a clean, interpretable feature for classification (e.g., peak rate, area under the curve).

Sum of Firing Rate over Time:

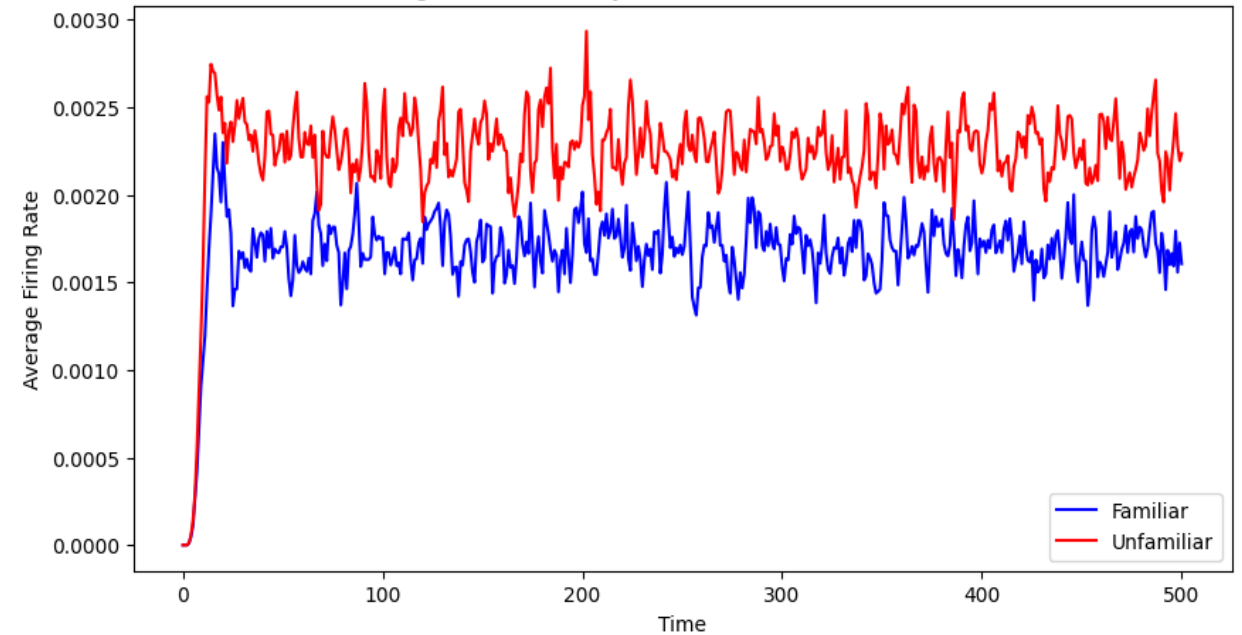
- Focuses on the total activity, less on spike timing.
- Simplifies to a scalar feature per neuron or trial for classification.
- Compares overall responses to familiar vs. unfamiliar stimuli.

Visualizing Firing Rate over Time

Sum of Neural Activity for Familiar vs Unfamiliar Stimuli



Average Neural Activity for Familiar vs Unfamiliar Stimuli



Unfamiliar stimuli elicit a more prolonged response across the neuron population

Data Preprocessing & Feature Extraction

Neural Coding Methods:

- **Mean:** average firing rate per neuron over time
- **Sum:** total firing per time step (collapse neuron axis)
- **Fourier:** frequency-domain signal magnitude
- **PCA:** dimensionality reduction over neurons $\times$ time
- **Justification:** Each method captures different aspects of spike dynamics (e.g., rate coding, spectral features, global structure).

Fourier

- **Purpose:** Converts time-domain signals into frequency-domain representations.
- **How it works:**
 - Breaks down a signal into its constituent frequencies, capturing periodic and oscillatory patterns.
 - Provides insight into the rhythm and periodicity of neural responses.
- **Limitations:**
 - Assumes periodicity, which may not always match the complexity of biological signals.
 - May not capture transient, non-periodic events well.

PCA

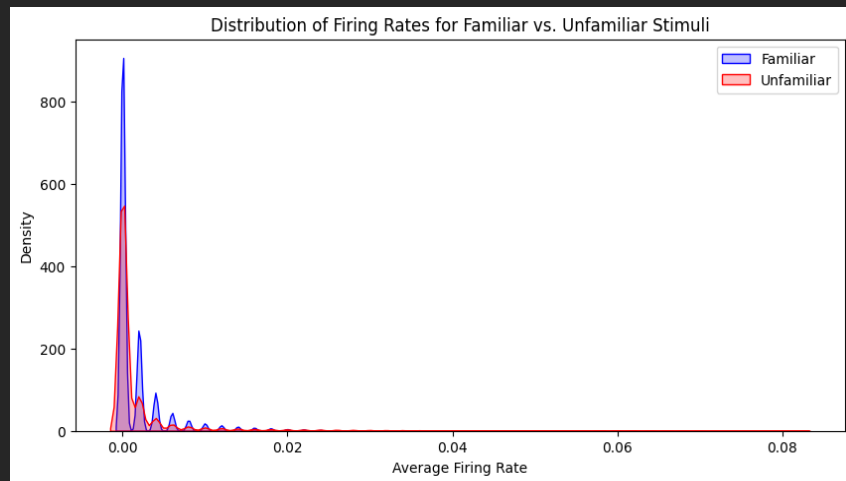
- **Purpose:** Dimensionality reduction technique that identifies the principal components of data.
- **How it works:**
 - Converts high-dimensional data into a smaller set of components (principal components) that capture the maximum variance.
 - Reduces the number of variables while retaining most of the information.
- **Limitations:**
 - May lose important class-specific information when reducing dimensions too much.
 - Works well when data is linearly separable but may struggle with non-linear relationships.

Exploratory Analysis

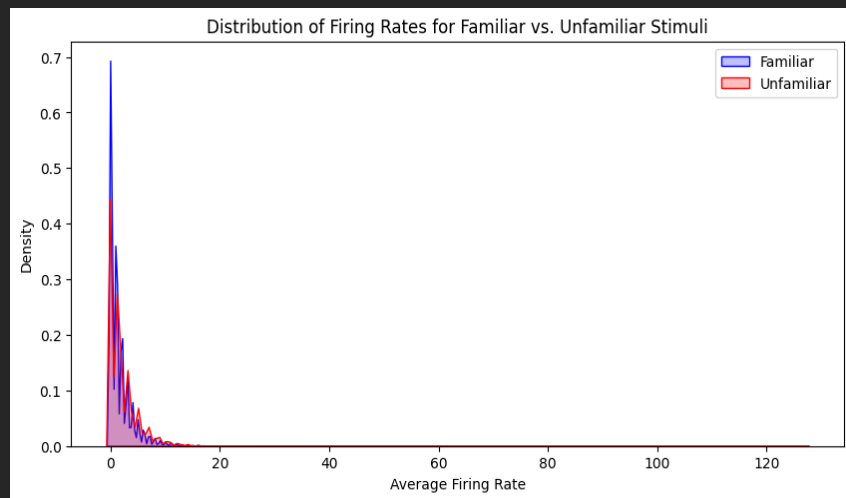
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- This function visualizes neural feature distributions for familiar and unfamiliar stimuli using kernel density estimation (KDE).
 - Assesses class separability: Are the distributions distinct or overlapping?
 - Supports feature transformation choices and justification.
 - Lays the foundation for quantitative comparisons (e.g., KL divergence, Wasserstein distance).

Exploratory Analysis

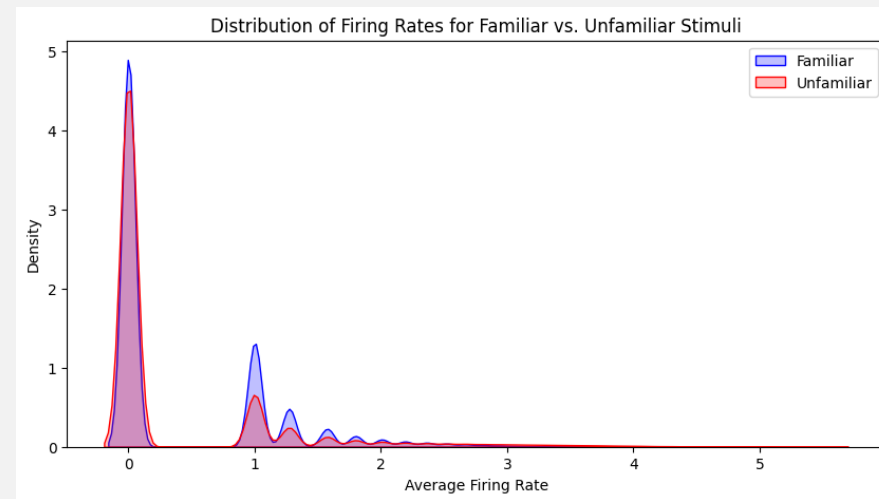
Mean:



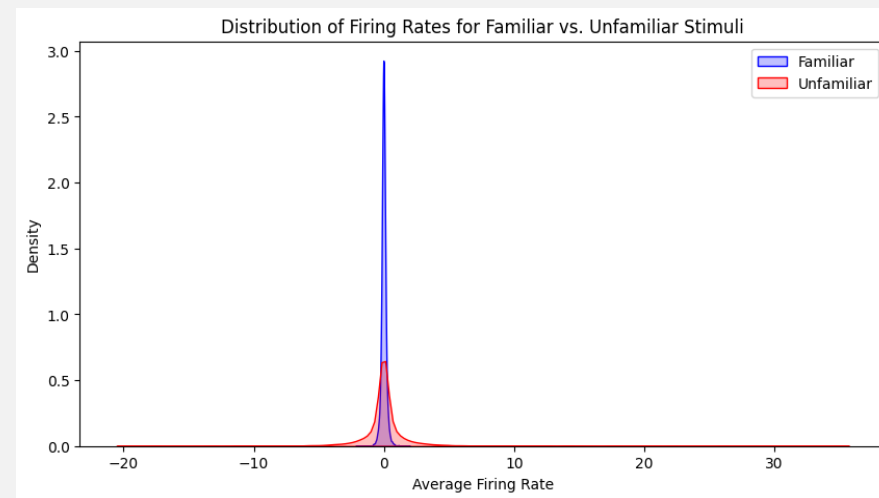
Sum:



Fourier:



PCA:



Wasserstein Distance:

- Measures the "distance" between two distributions by considering the minimal cost to transform one into the other.
- More robust to differences in distribution support and outliers.
- Provides a smoother, less sensitive measure than KL divergence in some cases.
- **PCA** (0.82) and **Fourier** (0.15) show moderate to low distance, indicating some level of separation between distributions for these features.
- **Sum** (0.56) shows a larger distance, indicating less overlap, while **Mean** (0.001) has the smallest distance, suggesting a strong overlap between familiar and unfamiliar stimuli.

KL Divergence:

- Measures the difference between two probability distributions.
- Sensitive to distribution shape; works well for distributions with similar support.
- Penalizes differences in areas with zero probability in one distribution.
- Higher values for **PCA** (5.08) and **Fourier** (1.58) suggest larger differences between the distributions for these transformations.
- **Mean** and **Sum** show lower divergence, indicating more similar distributions for familiar and unfamiliar stimuli in these cases.

Confusion Matrix

Classification Method: Logistic Regression

Chosen for its simplicity and interpretability in binary classification (familiar vs unfamiliar).

Data Splitting:

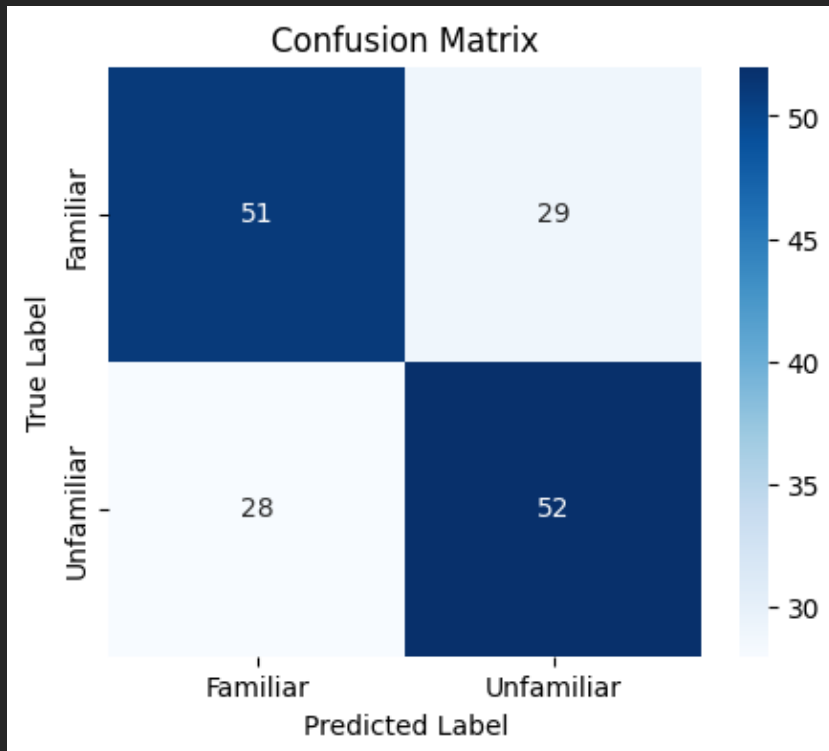
- 80% training, 20% testing (stratified to preserve class ratio)

Purpose:

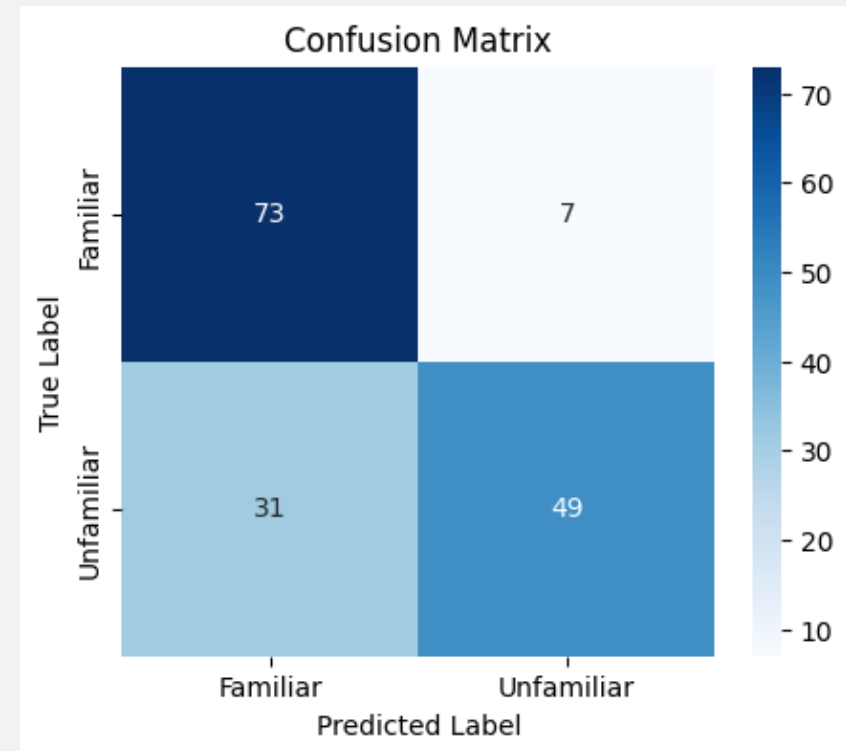
- Helps assess model performance beyond overall accuracy.
- Identifies bias in prediction toward one class.

Confusion Matrix

Sum:

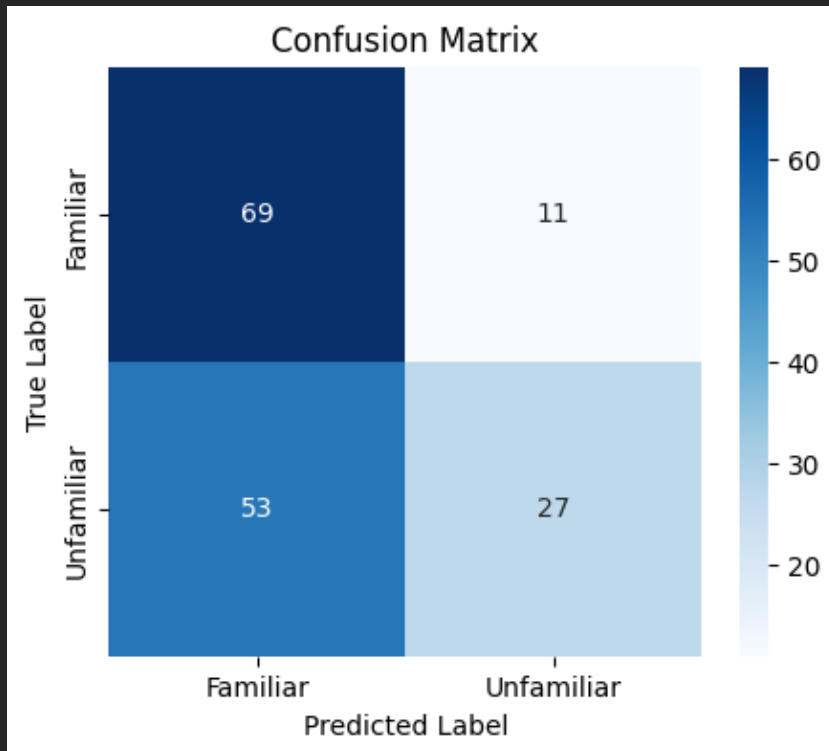


Mean:

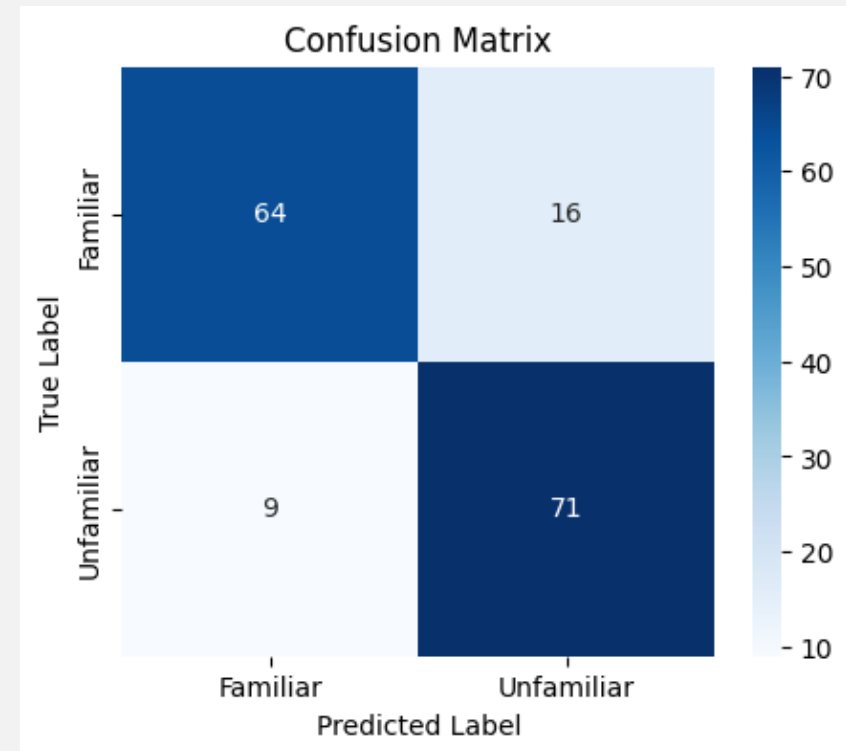


Confusion Matrix

PCA:



Fourier:



Confusion Matrix

- **Fourier features** produced the most balanced and effective classifier.
-> Reduces both false positives and false negatives.
- **Mean features** provided a good balance between precision and recall, with relatively high accuracy and few false positives.
- **PCA features** resulted in poor recall, indicating many familiar stimuli were misclassified as unfamiliar.
- **Sum features** showed moderate balance but lower overall performance, with more false positives than other methods.

Accuracy & Precision

Accuracy:

- **Definition:** The proportion of correctly classified samples out of all samples.
- **Formula:** $(TP+TN)/(TP+TN+FP+FN)$
- **When to use:** Useful when the classes are balanced. May be misleading in imbalanced datasets.

Precision:

- **Definition:** The proportion of correctly predicted positive samples out of all predicted positive samples.
- **Formula:** $TP/(TP+FP)$
- **When to use:** Important when the cost of false positives is high (e.g., diagnosing a disease).

Recall & F1-score

Recall:

- **Definition:** The proportion of correctly predicted positive samples out of all actual positive samples.
- **Formula:** $TP / (TP + FN)$
- **When to use:** Crucial when the cost of false negatives is high (e.g., missing a rare event).

F1-score:

- **Definition:** The harmonic mean of Precision and Recall, balancing both metrics.
- **Formula:** $2 \cdot ((Precision \cdot Recall) / (Precision + Recall))$
- **When to use:** Useful when there is a need to balance Precision and Recall, particularly in imbalanced datasets.

Method	Accuracy	Precision	Recall	F1-Score
Mean	0.7625	0.8750	0.6125	0.7206
Sum	0.6438	0.6420	0.6500	0.6460
Fourier	0.8438	0.8161	0.8875	0.8503
PCA	0.6250	0.7778	0.3500	0.4828

Evaluating model performance

✓ **Best Performance:** Fourier

✗ **Worst Generalization:** PCA (despite high KL)

Evaluating model performance

- **Accuracy:** Ranges from 62.5% to 84.38%. High accuracy in **Fourier** suggests overall good performance, but accuracy alone may overlook imbalances in class distribution.
- **Precision:** Higher in **Fourier** and **Mean** (0.82, 0.88). Indicates the model is effective at minimizing false positives, particularly in identifying **Unfamiliar** stimuli.
- **Recall:** Highest in **Fourier** (0.89), meaning the model detects most **Unfamiliar** stimuli. **PCA** has the lowest recall (0.35), missing many positives.
- **F1-score:** Best in **Fourier** (0.85), balancing precision and recall well. **PCA** has the lowest F1-score (0.48), reflecting poor performance in both precision and recall.